UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SECI1143 – KEBARANGKALIAN STATISTIK & ANALISIS DATA (PROBABILITY & STATISTICAL DATA ANALYSIS)

SECTION 03

ASSIGNMENT 1 – CHAPTER 1 &2

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FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject	:	PROBABILITY & STATISTICAL DATA ANALYSIS (SECI / SCSI 2143/ SECI 1143)
Task	:	Chapter 1 & Chapter 2

INSTRUCTION:

1. This is a **GROUP** assignment. Please clearly write the group members name & matric number in the front page of the submission.
2. This assignment contributes to 5% of overall course marks.
3. Only **HANDWRITTEN** submission is accepted:
   a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc,) will be **REJECTED**.
   b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
   c. For answer that need to draw graphs, using graph paper is optional. You can use plain paper.
   d. Round your answers to **TWO** decimal places.
   e. Please scan/snapshot your work and save as a PDF file.
4. Submission via eLearning – only **ONE** group member needs to submit on behalf of the group.

QUESTION 1[10 MARKS]

You are a research assistant at a university's Faculty of Computing. The faculty is conducting a study to explore the relationship between faculty students' socioeconomic backgrounds and their academic performance. The aim is to identify potential factors that may influence academic success and provide insights for educational policymakers and practitioners. Identify the following:

a) What is population and sample?
b) List possible variables/data to be collected and identify characteristic of the variables/data according to the following
   i. Discrete/continuous
   ii. Categorical/numerical

iii. Level of measurement: nominal, ordinal, ratio or interval

Give example of the sample of data to help your justification

c) Primary Data Source

d) Secondary data source

e) Identify and construct possible descriptive statistics that can be used to summarized the data. Note : You only need to put any random figures in the constructed descriptive statistics.

QUESTION 2[10 MARKS]

A student trainer worked in the training room in the physical education building. The number of students coming in the taped or for treatment of other injuries was recorded each half hour, yielding the following data. Calculate the frequency and relative frequency of the data

3	3	2	0	4	5	6	4	4	3
2	1	2	3	0	5	5	3	2	3
5	4	1	2	0	3	2	4	2	6

QUESTION 3 [25 MARKS]

A packaged product has a label weight of 450 grams. The company goal is to fill each packages with at least 450 grams, but at most 458 grams. To check these goals , a random sample of 100 packages was selected and weighted, yielding the following weights, rounded to the nearest gram:

450	**450**	458	460	**450**	472	455	462	463	467
453	455	466	453	471	**451**	472	464	463	453
469	457	460	458	472	454	473	466	455	470
470	472	**452**	459	453	466	457	464	462	459
472	**450**	471	464	453	472	457	461	**450**	470
471	467	456	465	**450**	**450**	457	465	466	454
472	467	455	456	455	470	471	471	**450**	458
469	467	457	455	463	453	458	**452**	456	469
457	472	460	459	471	461	470	453	469	471
450	454	**451**	**450**	456	472	465	464	471	467

Construct :

a) Construct a histogram,
b) frequency polygon
c) ogive

QUESTION 4 [10 MARKS]

Below are the lifetime, in hours of fifty 40-watt 110-volt internally frosted incandescent lamps, taken from forced life tests :

919	1196	78	1126	936	918
1156	920	948	1067	1092	1162
1170	929	950	905	972	1035
1045	855	1195	1195	1340	1122
938	970	1237	956	1102	1157
978	832	1009	1157	1151	1009
765	958	902	1022	1333	811
1217	1085	896	958	1311	1037
702	923				

Construct a modified box-plot and determine the outliers (if any)

Subject:

No:

Date:

1.(a) Population is the entire collection of individuals or object about which information is desired. Regarding to this study, population refer to the entire group of faculty students at the university's Faculty of Computing.

Sample is a subset of the population, selected for the study. For instance, the selected students from each year for the survey.

(b) (i) Discrete data - socioeconomic background. For example, the household size of a student's family, 1 person, 2 people or 3 people.

Continuous data - academic performance. For example, the GPA (Grade Point Average) of faculty students.

(ii) Categorical data - socioeconomic background. For example, the religious affiliation of faculty students (Muslim, Buddhist, Christian, Hindu)

Numerical data - academic performance. For example, the age of students, the CGPA of faculty students.

(iii) Nominal data - Religious affiliation of faculty students, such as Muslim, Buddhist, Christian and Hindu.

Ordinal data - Socioeconomic status including lower class, middle class, upper class.

Interval data - GPA of faculty students. For example, range from 0.0 to 4.0.

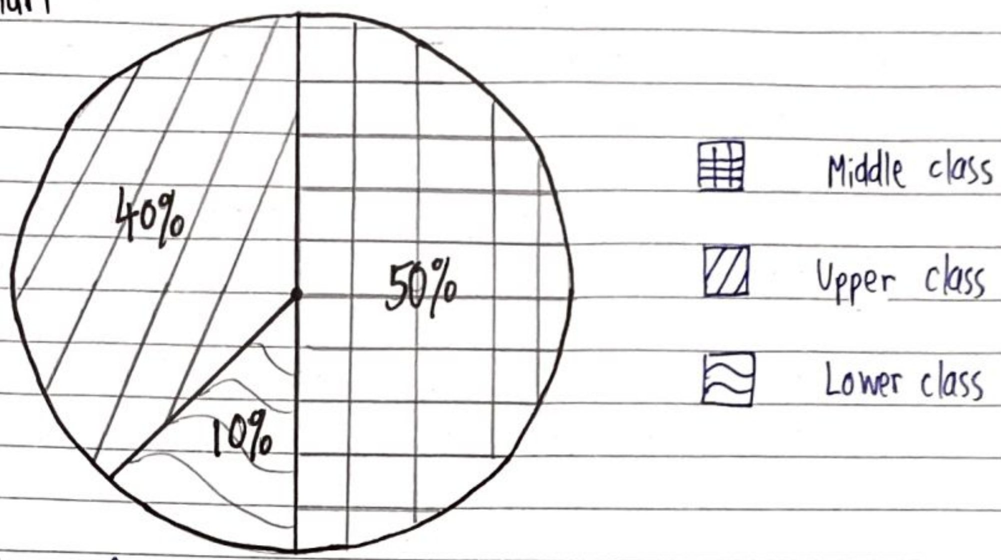
Ratio data - Annual family income. For example, RM5000, RM10000, RM20000.

(c) Primary data source - Surveys, questionnaires, interviews, and direct observations conducted specifically for this study to gather information from the students.

(d) Secondary data source - University records, academic performance databases, socioeconomic data from governmental or institutional reports that were collected for other purposes but can be used for this study.

Subject: .... No: .... Date:

(e) Pie chart



Percentage of Socioeconomic status for faculty students

Question 2

Number of students	Frequency	Relative Frequency
0	3	0.10
1	2	0.07
2	7	0.23
3	7	0.23
4	5	0.17
5	4	0.13
6	2	0.07
Total	30	1.00

Question 3

Weight	Tabulation	Frequency	Weight	Tabulation	Frequency
450	卌 卌	10	463	|||	3
451	||	2	464	||||	4
452	||	2	465	|||	3
453	卌 ||	7	466	||||	4
454	|||	3	467	|||||	5
455	卌 ||	6	468		0
456	||||	4	469	||||	4
457	卌 ||	6	470	卌	5
458	||||	4	471	卌 |||	8
459	|||	3	472	卌 ||||	9
460	|||	3	473	|	1
461	||	2			
462	||	2			

Range $= X_h - X_l$

$= 473 - 450$

$= 23$

cell interval, i

$$= \frac{R}{1 + 3.322 \log n}$$

$$= \frac{23}{1+3.322 \log 100}$$

$= 3.00$

number of cells, h

if $i = 3.00$

$$h = \frac{R}{i} = \frac{23}{3} = 7.67 \approx 8$$

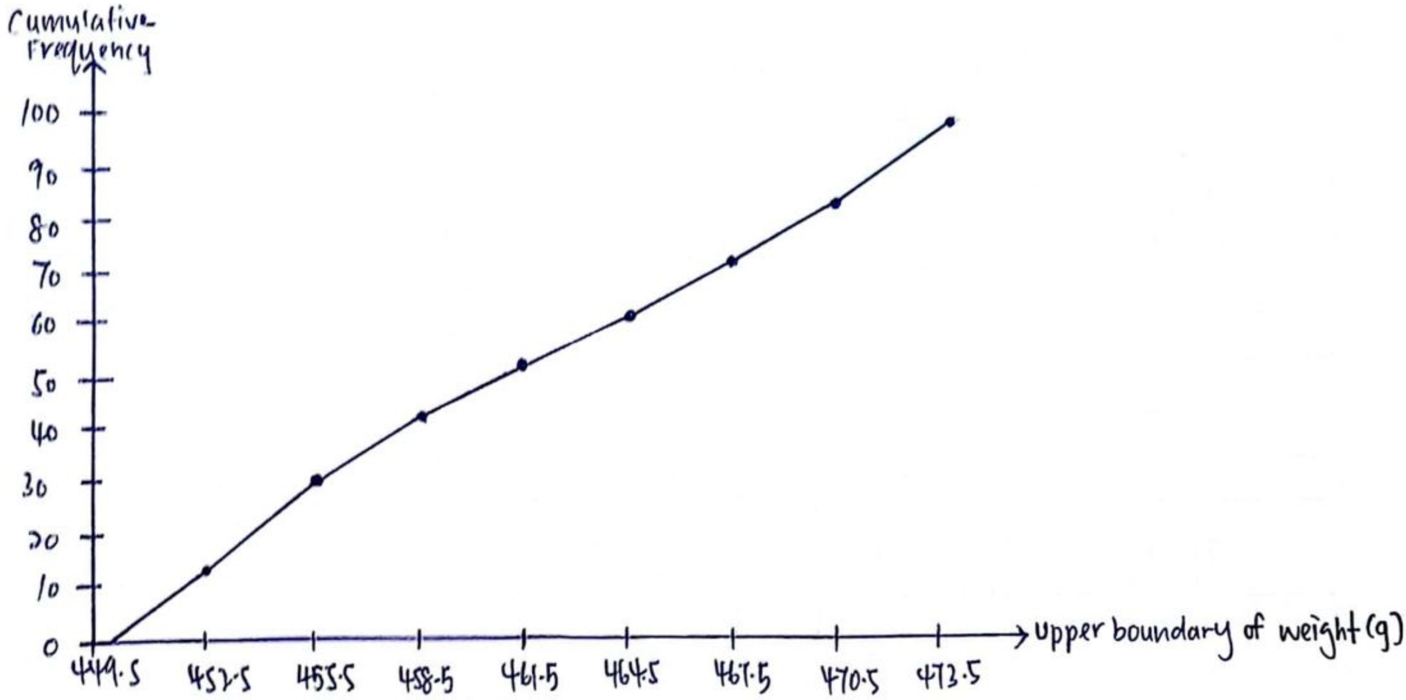
Midpoint

$$MP_1 = X_1 + \frac{i}{2}$$

$$= 450 + \frac{3}{2}$$

$= 451$

Ogive

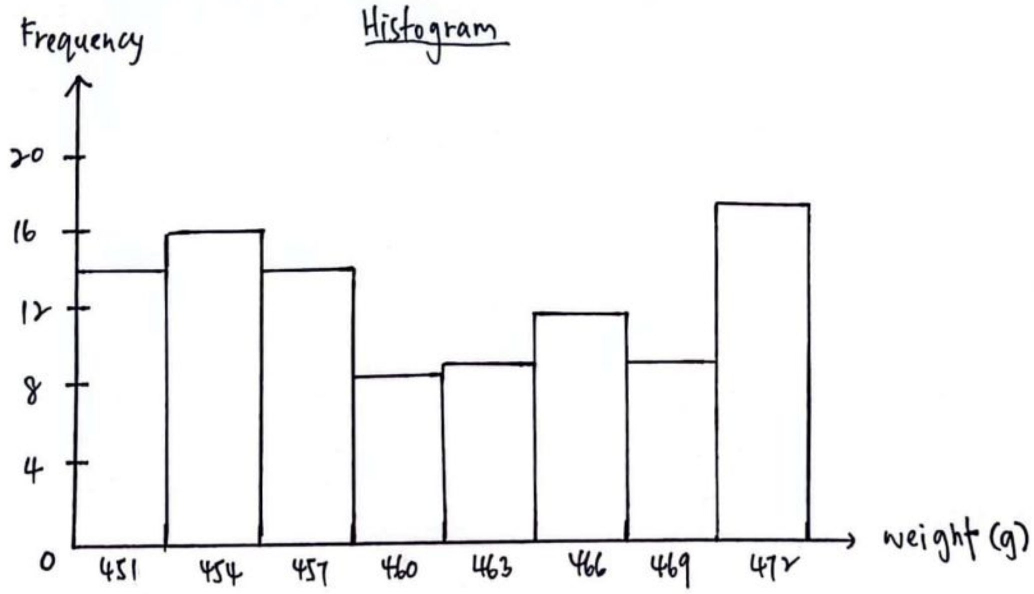


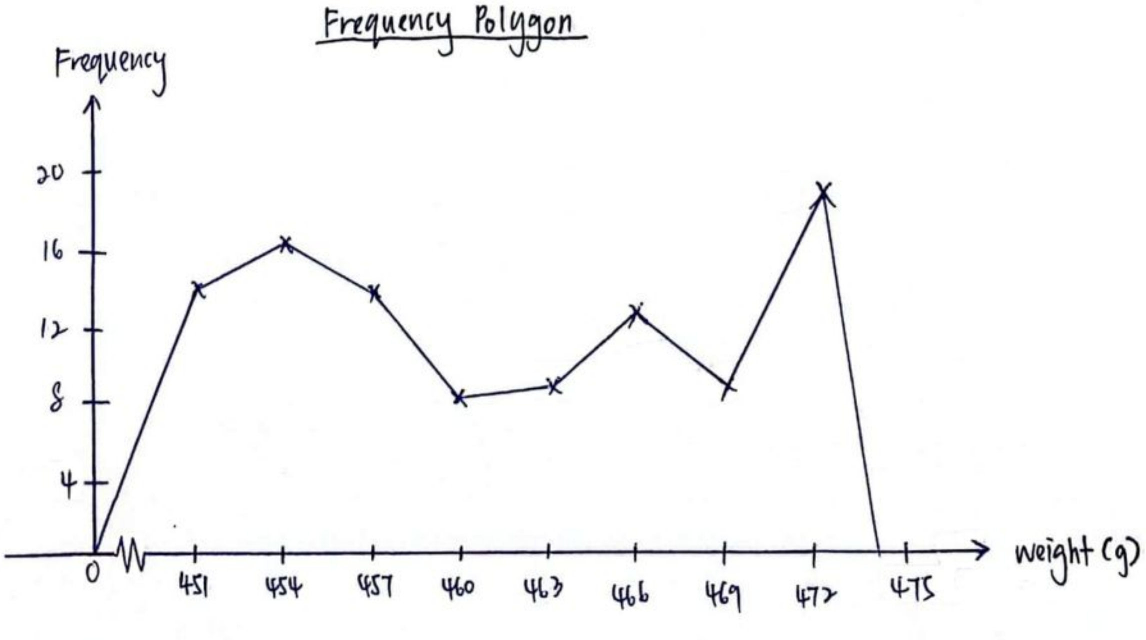
Cumulative frequency
100
90
80
70
60
50
40
30
20
10
0
449.5
452.5
455.5
458.5
461.5
464.5
467.5
470.5
473.5
upper boundary of weight (g)

Cell boundaries	Cell midpoint	Frequency	Cumulative Frequency
449.5 – 452.5	451	14	14
452.5 – 455.5	454	16	30
455.5 – 458.5	457	14	44
458.5 – 461.5	460	8	52
461.5 – 464.5	463	9	61
464.5 – 467.5	466	12	73
467.5 – 470.5	469	9	82
470.5 – 473.5	472	18	100

Boundaries:

$\frac{3.00}{2} = 1.50$





4. $N = 50$

$i = \frac{1}{4}(50)$

$= 12.5$

$k = 13$

$Y[13] = 923$

$Q_1 = 923$

$i = \frac{1}{2}(50)$

$= 25$

$\frac{Y[25] + Y[26]}{2}$

$= \frac{1009 + 1009}{2}$

$= 1009$

$Q_2 = 1009$

$i = \frac{3}{4}(50)$

$= 37.5$

$k = 38$

$Y[38] = 1156$

$Q_3 = 1156$

Lower boundary $= 923 - 1.5(1156 - 923)$

$= 573.5$

Upper boundary $= 1156 + 1.5(1156 - 923)$

$= 1505.5$

Outlier = 78